An echo from the past - building a Doppler repository for 'big data' in diving research

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BACKGROUND

DCS — **Decompression sickness** (DCS) remains a major operational concern for diving (naval, professional, and recreational)

VENOUS GAS EMBOLI — Historic table tests: 'dived out' using DCS as an endpoint. Today, gas bubbles/**venous gas emboli** (VGE), detected with ultrasound post-dive are often used as a **marker of decompression stress** in humans

DOPPLER US — **Doppler ultrasound (US)** is used to assess bubble loads. Grading of the bubbles is undertaken by well-trained specialists to try and reduce subjectivity

AIMS — To curate a large, open access database of Doppler recordings. In addition, it would be of benefit to develop an automated bubble monitoring system, based on **machine learning algorithms**

DIVING DATA AND ITS USES

HISTORIC DATABANK — The field of diving medicine has a historic bank of Doppler recordings from many past studies. These could be congregated and annotated - e.g. DCIEM (currently DRDC Toronto) table testing, NEDU table/equipment testing, QinetiQ dive data

Doppler data published in an accessible repository can facilitate:

- **Safeguarding of historic studies of immense scientific value**
- **Aid with development of robust automated monitoring (machine learning) system**
- **Enable 'big data' research approaches**
- **'New' insights with improved analytical techniques**
- **Increased power**

MATERIALS AND METHODS

BIG DATA → **TRAIN** → **TEST** → **VALIDATE**

The development of **machine learning algorithms** requires well-curated data sets for training, testing and validation

- **Large repositories** brought together by research collaborations are important
- They allow collection of enough data to allow **meaningful pooling** and ensure **high quality**
- Of **importance where data is relatively scarce** and grading requires specialist knowledge, e.g. Doppler monitoring

- Funding agencies already recognize and support these efforts
- For example the NIH has recently released 100,000 chest x-rays with annotated radiology reports, open for all
- **Aim is to collect the deidentified Doppler data for publication in an accessible repository**

PRINCIPLES OF ULTRASOUND FOR BUBBLE DETECTION

DOPPLER ULTRASOUND

- Transducer emits specific frequency signal (precordial 2-2.5 MHz)
- Receives a signal whose frequency has been Doppler shifted by moving objects or surfaces, such as red blood cells, valves etc. Bubbles are very good reflectors
- Operator hears the resulting difference in frequency
- Output is complex, coming from many different sources

BUBBLE LOADS

- The correlation between venous bubbles and DCS is non-linear
- Bubbles are a sensitive but non-specific predictor of DCS
- Conservative profiles generally give lower risk of DCS, extreme dives give higher risk of DCS; this is reflected by bubble loads and their correlation with DCS

KM GRADING

- Kisman-Masurel Code - Developed by Ken Kisman, DCIEM scientist, and Gerard Masurel, French Navy scientist (1978)
- Bubble signals are evaluated on multiple parameters, then combined and converted to give a semi-quantitative ordinal, non-parametric grade (Roman numerals). For example, KM code 222 gives a KM grade II
- 3 factors to grade the bubble signal: Frequency, Percentage/duration, Amplitude

RESULTS

THE DATABASE CONSISTS OF:

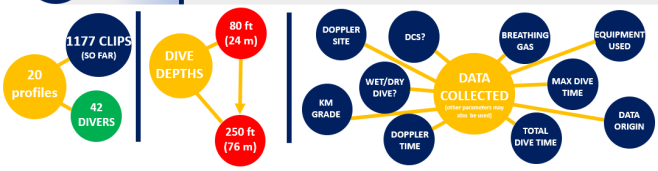
Audio files (.flac)

- The audio files are digitised Doppler recordings drawn from original analogue audio cassettes and DAT tapes
- Cut into individual measurements, or 'clips', e.g. precordial at rest, left subclavian at flex

Associated 'metadata'

- The metadata is the annotation relating to the specific Doppler recording
- Headed by an unique filename
- All inputs are standardised for machine reading

FIRST DATA SET — Audio C90 tapes digitised – data source: NEDU air algorithm trials performed in the 1980s. The data extracted so far consist of:



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE
1	Doppler bubble rep	itory da	ase. Th	colum	n red ar	nodifier	o the previous colu	n																							
2	Data collected																Change to minutes?						post dive		sea water / fresh water						
3	filename	Site (subc	SC L or R	Rest or fle	KM Code	KM grade	Dive max depth	m or ft	gas	Nitrogen I	Oxygen PI	Helium PF	ATA or kP	time to m	bottom tia	ascent tir	Total dive	Age (year	Sex	BMI	Data origi	DCS? (y/n	Type of DI	Doppler tim	Chamber/	sw or fw	Study ID	Subject ID	type of sc	repetitive dive (y/n	
4	THe3005_00001.flac	PC		R		0	120	ft	Heliox		0.7	3.85	ATA				00:40	32	M		Thalmann n			7	wet	fw	HE3005	152	Techno Scientific		
5	THe3005_00002.flac	PC		F		0	120	ft	Heliox		0.7	3.85	ATA				00:40	32	M		Thalmann n			7	wet	fw	HE3005	152	Techno Scientific		
6	THe3005_00003.flac	PC		F		0	120	ft	Heliox		0.7	3.85	ATA				00:40	32	M		Thalmann n			7	wet	fw	HE3005	152	Techno Scientific		
7	THe3005_00004.flac	PC		F		0	120	ft	Heliox		0.7	3.85	ATA				00:40	32	M		Thalmann n			7	wet	fw	HE3005	152	Techno Scientific		
8	THe3005_00005.flac	SC	R	R		0	120	ft	Heliox		0.7	3.85	ATA				00:40	32	M		Thalmann n			7	wet	fw	HE3005	152	Techno Scientific		
9	THe3005_00006.flac	SC	R	F		0	120	ft	Heliox		0.7	3.85	ATA				00:40	32	M		Thalmann n			7	wet	fw	HE3005	152	Techno Scientific		
10	THe3005_00007.flac	SC	R	F		0	120	ft	Heliox		0.7	3.85	ATA				00:40	32	M		Thalmann n			7	wet	fw	HE3005	152	Techno Scientific		

SUMMARY

CURRENT/FUTURE SOURCES OF POST-DIVE AUDIO DATA

- **Duke University, North Carolina**, data from historic trials including:
  - flying after diving, HeO_2, SIO_2, and NASA nucleation trials
- **Divers Alert Network (DAN) US**
  - Project Dive Explorer
- **QinetiQ, UK**
  - HeO_2 dives, table testing, submarine saturation and escape dives, animal model dives
- **Defence Research and Development Canada (DRDC), Toronto Research Centre (formerly DCIEM)**
  - Historic DCIEM table testing – 'Sawatzky' data
- **Coordination of project by UC San Diego**

We would welcome any further collaborations: Please contact plindholm@health.ucsd.edu

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